

Fig. 7: Concept of beam rider guidance

homing guidance and re-transmission homing guidance.

Navigation guidance further comprises: inertial navigation guidance, ranging navigation guidance, celestial navigation guidance and geophysical navigation.

Beam rider guidance.

Beam-riding guidance of munitions is based on a radar beam or a laser beam, constantly pointed toward the target throughout the flight time of the munitions. After the missile is launched, it attempts to keep itself inside the beam, while the aiming station keeps the beam pointed at the target.

The missile's flight path control functions as follows: The missile's guidance sensors located at the rear of the missile receive information about the position of the missile within the beam. The missile interprets this information and generates its own correction signals. These correction signals are used to send command signals to the control surfaces of the weapon to keep the missile in the centre of the beam. The launch station keeps the beam pointed at the target throughout the engagement period and the missile rides the beam to the intended target. Both radar and laser beam rider guidance

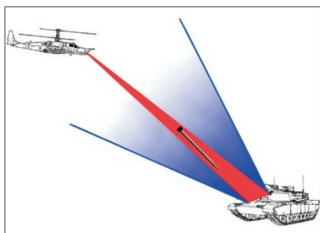


Fig. 8: Concept of laser beam rider

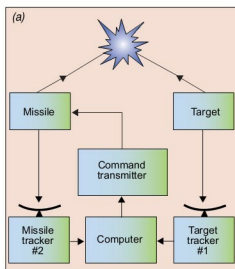
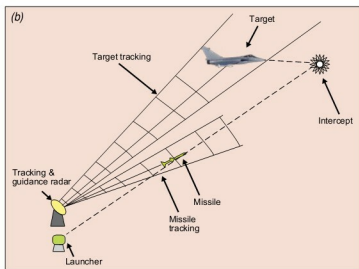


Fig. 9: Command guidance



have been successfully employed for surface-to-surface, surface-to-air and air-to-ground weapons.

Fig. 7 illustrates the concept of beam riding for a surface-to-air weapon using a radar beam. Fig. 8 shows a laser beam rider missile launched from a helicopter against a tank target. As the beam moves farther away from the launcher and towards the target, it spreads out and it becomes difficult to keep the beam in the centre of the target. That makes the beam rider concept effective only for short to medium operational ranges.

Laser beam riding guidance became more attractive particularly for short-range anti-air and anti-tank missiles in the 1980s and 1990s with the introduction of low-cost and highly-portable laser designators. Laser beam riding also allows the designer to

encode additional information in the beam using digital means. Laser beam rider missiles are inherently more accurate. Also, a narrow laser beam guiding the weapon makes it more difficult to be detected and therefore immune to countermeasures.

Command guidance. In this case, the missile is commanded on an intercept course with the target. Conventionally, this is achieved by using two separate radars to continuously track the target and missile. Tracking data from these radars is fed to a computer that computes the trajectories of the two vehicles. The computer, in turn, sends appropriate command signals over a radio link to the missile. A sensor onboard the missile decodes the commands and operates the control surfaces of the missile to adjust its course so as to intercept the target in flight.